Question			Answer	Marks	AO	Guidance
14			$5 - \cos\theta - 6(1 - \cos^2\theta) = 0$	**B1**	**3.1a**	Using the correct identity. May be seen anywhere in the working. Condone poor notation such as missing arguments.
			$6\cos^2\theta - \cos\theta - 1 = 0$	**M1**	**2.1**	Simplifies to **3 term** quadratic **in cosine** and sets equal to 0. Can score this mark if $1 + \cos^2\theta$ used (gives an unsolvable quadratic).
			eg $(3\cos\theta + 1)(2\cos\theta - 1) = 0$	**M1**	**1.1**	Attempt to solve a quadratic in cosine- may see use of quadratic formula; allow sign errors only in either method. **Calculator methods not acceptable here- detailed reasoning required.** If using QF a correctly quoted formula followed by a slip in the substitution of values scores M1, but if the formula isn't quoted and there are errors in the substitution then M0.
			$\cos\theta = \frac{1}{2}$ ***and*** $\cos\theta = -\frac{1}{3}$ seen	**A1**	**1.1**	
			$\underline{60°}$ ***and*** $\underline{° 300°}$ **or** $\underline{109°}$***and*** $\underline{251°}$ **or** $60°$***and*** $\underline{109°}$	**A1**	**1.1**	A complete solve for either trig equation or two 'first values' obtained: allow 109.47… and 250.53…; or e.g. 109.5 and 250.5 (or awrt 109 and 251)

Question			Answer	Marks	AO	Guidance
			all 4 correct with no extras	**A1**	**1.1**	SC 1: If the candidate uses a calculator to obtain $\cos\theta = \frac{1}{2}$ ***and*** $\cos\theta = -\frac{1}{3}$ then proceeds to a fully correct solution, then award SC 4/6 SC 2: If the candidate uses a calculator to obtain $\cos\theta = \frac{1}{2}$ ***and*** $\cos\theta = -\frac{1}{3}$ then proceeds to partially correct solution with at least two correct values then award SC 3/6
				[6]		

Question	Answer	Marks	AO	Guidance